# Pointwise Convergence Is the Least Locally Solid Hausdorff Topology

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## Status

This packet records a candidate full solution, likely valid, to Question 6.c.1 of Kiwerski and Kolwicz [1]: whether local convergence in measure is the coarsest locally solid Hausdorff topology in the class of Banach sequence spaces.

## Source Passage

> quence space. At the heart of the proof of Remark 5.a.4 is the observation that the minimal Hausdorff topology on $X$ coincide with the topology of point-wise convergence when restricted to the unit ball $\mathrm{Ball}(X)$. Since the topology of point-wise convergence on $X$ coincide with the topology of local convergence in measure, so it is natural to ask the following
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> **Question 6.c.1.** *Is the topology of local convergence in measure the coarsest locally solid Hausdorff topology in the class of Banach sequence spaces?*
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> Note that in 1987 Labuda showed that the topology of local convergence in measure is the coarsest locally solid topology in the class of Orlicz spaces (see [Lab87, Theorem 14]).
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> 6.d. **Glimpse of Kalton's Zone.** It is relatively easy to notice significant gaps in knowledge regarding Kadets–Klee properties in the world of *quasi-Banach spaces*. There are actually good reasons for this. For example, as Day's classical result shows, the topologi-

## Transcription of the Open Question

The source paper asks:

> "Is the topology of local convergence in measure the coarsest locally solid Hausdorff topology in the class of Banach sequence spaces?"

Immediately before the question, the authors note that for a Banach sequence space the topology of pointwise convergence coincides with local convergence in measure. They also note Labuda's 1987 theorem for Orlicz spaces [2].

## Definitions Used

Let $\Gamma$ be countable and equipped with counting measure. Following [1], a Banach sequence space $X$ over $\Gamma$ is a Banach space which is a linear subspace of $\omega(\Gamma)$, contains $\mathbf{1}_F$ for every finite $F \subseteq \Gamma$, and has the ideal property: if $|x(\gamma)| \leq |y(\gamma)|$ for all $\gamma \in \Gamma$ and $y \in X$, then $x \in X$.

For $\gamma \in \Gamma$, write $e_\gamma = \mathbf{1}_{\{\gamma\}}$. Let $\tau_{\text{pt}}$ denote the topology of pointwise convergence on $X$, that is, the initial topology generated by the coordinate maps

$$\pi_\gamma : X \to \mathbb{R}, \qquad \pi_\gamma(x) = x(\gamma).$$

A subset $U \subseteq X$ is solid if $x \in U$, $y \in X$, and $|y(\gamma)| \leq |x(\gamma)|$ for all $\gamma$ imply $y \in U$. A vector topology on $X$ is locally solid if zero has a base of solid neighborhoods.

## Theorem

**Theorem 1.** *Let $X$ be a Banach sequence space over a countable set $\Gamma$. Then the pointwise topology $\tau_{\text{pt}}$ is the least Hausdorff locally solid vector topology on $X$. Consequently, for the counting measure on $\Gamma$, the topology of local convergence in measure is the coarsest locally solid Hausdorff topology in the class of Banach sequence spaces.*

## Proof Intuition

The only possible way for a Hausdorff vector topology to be too coarse is for it not to see some coordinate. Solidity prevents this. If a solid neighborhood contains $x$, then it also contains the one-coordinate fragment $x(\gamma)e_\gamma$. Hausdorffness on the one-dimensional coordinate axis forces these fragments to have small scalar coefficient inside sufficiently small neighborhoods. Hence every coordinate map is continuous for every Hausdorff locally solid topology.

## Proof

First observe that $\tau_{\text{pt}}$ is itself a Hausdorff locally solid vector topology. Indeed, a base at zero is given by the finite-coordinate sets

$$U(F, \varepsilon) = \{x \in X : |x(\gamma)| < \varepsilon \text{ for all } \gamma \in F\},$$

where $F \subseteq \Gamma$ is finite and $\varepsilon > 0$. These sets are solid, and the resulting topology is Hausdorff because the coordinates separate points of $X \subseteq \omega(\Gamma)$.

Now let $\tau$ be any Hausdorff locally solid vector topology on $X$. We prove that $\tau_{\text{pt}} \subseteq \tau$. Fix $\gamma \in \Gamma$ and $\varepsilon > 0$. Since $e_\gamma \neq 0$ and $\tau$ is Hausdorff, there is a balanced $\tau$-neighborhood $N$ of zero such that $e_\gamma \notin N$. Put $U = \varepsilon N$. If $\lambda e_\gamma \in U$, then $(\lambda/\varepsilon)e_\gamma \in N$. Were $|\lambda| \geq \varepsilon$, balancedness of $N$ would imply

$$e_\gamma = \frac{\varepsilon}{\lambda}\left(\frac{\lambda}{\varepsilon}e_\gamma\right) \in N,$$

a contradiction. Hence

$$U \cap \mathbb{R}e_\gamma \subseteq \{\lambda e_\gamma : |\lambda| < \varepsilon\}.$$

Because $\tau$ is locally solid, choose a solid $\tau$-neighborhood $V \subseteq U$. If $x \in V$, then $x(\gamma)e_\gamma \in X$ and

$$|x(\gamma)e_\gamma(\eta)| \leq |x(\eta)| \qquad (\eta \in \Gamma).$$

By solidity, $x(\gamma)e_\gamma \in V \subseteq U$, so the previous paragraph gives $|x(\gamma)| < \varepsilon$. Thus

$$V \subseteq \{x \in X : |\pi_\gamma(x)| < \varepsilon\},$$

which proves that $\pi_\gamma$ is $\tau$-continuous at zero, hence continuous.

All coordinate maps are therefore $\tau$-continuous. Since $\tau_{\mathrm{pt}}$ is the initial topology generated by these coordinate maps, it follows that $\tau_{\mathrm{pt}} \subseteq \tau$. As $\tau$ was arbitrary, $\tau_{\mathrm{pt}}$ is the least Hausdorff locally solid vector topology on $X$.

Finally, for the counting measure on a countable set, local convergence in measure is convergence in measure on finite subsets of $\Gamma$. Since finite subsets carry counting measure with finitely many atoms, this is equivalent to coordinatewise convergence on every finite set, i.e. to $\tau_{\mathrm{pt}}$. The stated affirmative answer follows. □

## Verification Notes

No numerical verification is involved. The proof uses only the structural axioms in the source paper's definition of Banach sequence spaces: finite coordinate vectors belong to $X$, and the ideal property guarantees that coordinate fragments $x(\gamma)e_\gamma$ belong to $X$ and lie in solid neighborhoods whenever $x$ does.

The proof does not use the norm topology. It gives the stronger statement that pointwise convergence is below every Hausdorff locally solid vector topology on the underlying sequence lattice.

## Novelty and Literature Check

The run indexes were searched for arXiv:2411.09430, "Kadets", "Kadec", "Klee", "direct sums", "coarsest locally solid", "minimal locally solid", "local convergence in measure", and related phrases. No existing packet or attempt addresses this question.

External searches on 2026-06-19 for exact phrases from Question 6.c.1 and for "coarsest locally solid Hausdorff topology" with "Banach sequence space" found the source paper and adjacent minimal-topology literature, especially Kandic and Taylor [3], but no later work explicitly resolving this Banach sequence space question. Labuda's theorem [2], cited by the source paper, treats Orlicz spaces and is consistent with the argument here.

Human review should focus on the standard locally solid topology convention: zero has a base of solid neighborhoods. Under that convention, the coordinate-axis argument above is the complete proof.

## References


[1] T. Kiwerski and P. Kolwicz, *Direct sums and abstract Kadets–Klee properties*, arXiv:2411.09430.

[2] I. Labuda, *Submeasures and locally solid topologies on Riesz spaces*, Mathematische Zeitschrift 195 (1987), 179–196.

[3] M. Kandic and M. A. Taylor, *Metrizability of minimal and unbounded topologies*, arXiv:1709.05407.